Smoothing splines with some discontinuities

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Abstract

We generalize the class of cubic smoothing splines by allowing a finite number of discontinuities in unknown positions. Our approach is based on the deformation of the domain of the x variable.

1 Introduction

2 Methodology

To introduce sharp discontinuities within a smooth framework, we depart from the classical smoothing spline approach by introducing a dual-domain formulation. We distinguish between the physical domain, where observations occur, and a latent domain, where the function's smoothness is evaluated.

Physical vs. Latent Domains Let the observed data be pairs (x_i, y_i) for $i = 1, \dots, n$, where x_i represents the physical time (or spatial coordinate). The physical interval between consecutive knots is defined as $\delta_i = x_{i+1} - x_i$. We introduce a non-negative vector of latent variables, $\boldsymbol{\gamma} = \{\gamma_1, \dots, \gamma_{n-1}\}$, which we term "time jumps". These variables define a mapping from the physical domain to a dilated latent domain. The effective length of the i-th interval in the latent domain is given by:

$$\tau_i = \delta_i + \gamma_i$$

Here, γ_i represents the amount of *fictitious time* inserted between observations i and $i + 1$. If $\gamma_i = 0$, the physical and latent intervals are identical. If $\gamma_i > 0$, the latent interval is expanded in the segment $[x_i, x_{i+1})$.

The Functional Mapping The core of the method lies in the definition of the spline function $f_i(\cdot)$ over the i-th interval. Formally, $f_i(s)$ is a cubic polynomial defined on the latent coordinate $s \in [0, \tau_i]$. The relationship between the physical coordinate x (where $x_i \leq x \leq x_{i+1}$) and the latent coordinate s is governed by a linear dilation:

$$s_(x) = (x - x_i) \cdot \frac{\tau_i}{\Delta x_i} = (x - x_i)\left(1 + \frac{\gamma_i}{\Delta x_i}\right)$$

Consequently, the function observed in physical space, $f(x)$, is a projection of the latent function: for $i = 1, 2, \ldots, n-1$

$$f(x) \equiv f_i(s_i(x)) \quad \text{for } x_i \leq x < x_{i+1}.$$

The function $f_i(s)$ is smooth in the latent domain s. However, when projected back to the physical domain x, the dilation factor compresses this smoothness into a smaller interval, potentially resulting in a steep gradient (a "jump") if γ_i is large.

The Variational Objective Function The optimization problem seeks to minimize the functional $J(f, \boldsymbol{\gamma})$, which balances data fidelity and the smoothness of the latent function.

The objective function is defined as:

$$J(f, \boldsymbol{\gamma}) = \underbrace{\sum_{i=1}^{n} (y_i - f(x_i))^2}_{\text{Distance from observations}} + \lambda \underbrace{\sum_{i=1}^{n-1} \int_0^{\tau_i} \left(\frac{d^2 f_i(s)}{ds^2}\right)^2 ds}_{\text{Roughness in the dilatated domain}}$$

subject to $\gamma_i \geq 0$ and $\sum \gamma_i \leq M$ (budget constraint).

Interpretation of the Mechanism The efficacy of this formulation becomes apparent when analyzing the energy term. For a cubic spline segment $f_i(s)$ connecting two values with a height difference $\Delta y = y_{i+1} - y_i$, the minimum possible roughness energy is inversely proportional to the cube of the interval length:

$$\int_0^{\tau_i} (f_i''(s))^2 ds \propto \frac{(\Delta y)^2}{\tau_i^3} = \frac{(\Delta y)^2}{(\Delta x_i + \gamma_i)^3}$$

This relationship reveals the trade-off handled by the optimizer. In regions of continuity, the algorithm sets $\gamma_i \approx 0$. The latent interval equals the physical interval ($\tau_i \approx \Delta x_i$), and the model behaves like a standard cubic

smoothing spline. In regions of discontinuity (jumps), if $|\Delta y|$ is large while δ_i is small, the standard energy term would explode ($\to \infty$), forcing a standard spline to over-smooth. Instead, our algorithm allocates a large γ_i. This dilates the denominator $(\delta_i + \gamma_i)^3$, drastically reducing the penalty cost.

In essence, the variable γ_i allows the function to "borrow time" to complete a transition. While the transition appears instantaneous in physical time (a jump), it remains a smooth, low-energy curve in the latent time dimension. The parameter M acts as a sparsity constraint, ensuring that such time-warping occurs only where strictly necessary to fit the data.

3 Smoothing splines with jumps as a signal extraction problem

Thanks to the works of Wahba (1978) and Wecker and Ansley (1983) we know that smoothing splines can also be obtained through signal extraction techniques. In particular, the cubic smoothing spline is given by

$$\hat{f}_t = \mathbb{E}\left[f(x_t)|Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n\right], \tag{1}$$

where y_t is generated by the process

$$\begin{aligned} Y_t &= f(x_t) + \varepsilon_t, \quad \varepsilon_t \sim \text{NID}(0, \lambda\sigma^2) \\ \frac{df^2}{dx^2} &= \sigma w(x), \end{aligned}$$

with $w(x)$ a continuous-time standard white noise process and diffuse initial conditions for $f(x_1)$ and its derivative.

The above system can be cast in the following state-space form (see Harvey and Koopman, 2000, for example):

$$\begin{aligned} Y_t &= f(x_t) + \varepsilon_t, \quad \varepsilon_t \sim \text{NID}(0, \lambda\sigma^2) \\ \begin{bmatrix} f(x_{t+1}) \\ f'(x_{t+1}) \end{bmatrix} &= \begin{bmatrix} 1 & \delta_t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} f(x_t) \\ f'(x_t) \end{bmatrix} + \begin{bmatrix} \eta_t \\ \zeta_t \end{bmatrix}, \quad \begin{bmatrix} \eta_t \\ \zeta_t \end{bmatrix} \sim \text{NID}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \sigma^2 \begin{bmatrix} \frac{\delta_t^3}{3} & \frac{\delta_t^2}{2} \\ \frac{\delta_t^2}{2} & \delta_t \end{bmatrix} \right) \end{aligned}$$

with diffuse initial conditions:

$$\begin{bmatrix} f(x_1) \\ f'(x_1) \end{bmatrix} \sim \text{N}(\mathbf{0}, \kappa \boldsymbol{I}_2), \quad \text{with } \kappa \to \infty.$$

The Kalman filter applied to this form enables the numerical maximum likelihood estimation of the unknown parameters (σ, λ). Moreover, the state smoother provides the smoothing spline sequence of Equation (1).

It is straightforward to adapt this representation to the discontinuous case. Indeed, it is sufficient to substitute the physical time differences δ_t with their deformations $\tau_t = \delta_t + \gamma_t$. In this case, the likelihood function must be maximized with respect to $(\sigma, \lambda, \{\gamma_t\}_{t=1,\dots,n-1})$, respecting the constraints:

$$\gamma_t \geq 0, \qquad \sum_{t=1}^{n-1} \gamma_t \leq M.$$

An effective optimization algorithm for this problem is the one proposed by Svanberg (2002) and implemented in the *NLopt* library (Johnson, 2007). Since the likelihood function is not convex, good starting values for the sequence of γ_t are important: values that are close to zero are generally a good choice.

4 Alternating time-warped cubic spline algorithm

As an alternative to the numerical maximization of the likelihood, we propose a two-step algorithm with computational complexity $O(n)$ to carry out the computation of cubic smoothing splines with discontinuities.

This algorithm solves a variation of the cubic smoothing spline problem where the time intervals between observations are dynamic variables subject to optimization, allowing the function to "jump"or change regimes rapidly by warping the time domain.

The problem is non-convex due to the dependency of the integral on τ (and thus γ). We use *block coordinate descent*, alternating between two steps until convergence: the *Spline Step* (or f-step), in which we fix γ, solve for the optimal spline function f (a convex quadratic problem solvable via a linear system), and the *Warping Step* (or γ-step), in which, for fixed f, we solve for the optimal time jump vector γ. This is a constrained gradient descent problem.

The inputs of the algorithm are the following.

- n: Number of data points.
- x: Vector of size n (time points).
- y: Vector of size n (observations).
- λ: Smoothing parameter (scalar).
- M: Jump budget (scalar).

- η: Learning rate (scalar, e.g., 0.1).
- State Vector z: vector of size $2n$ containing function values and first derivatives:
$$z = [f_1, f_1', f_2, f_2', \ldots, f_n, f_n']^\top.$$

Step 1: The Spline Update (f-step) In this step, γ is fixed. We construct a linear system $(S + \lambda R)z = y^*$, where y^* is a vector of size $2n$ with y values at odd indices and 0 at even indices.

The Selection Matrix S is a diagonal matrix of size $2n \times 2n$. It acts as a mask, keeping the function values (indices $1, 3, 5 \ldots$) and zeroing out the derivatives (indices $2, 4, 6 \ldots$), because we only have data for f, not f'.

$$S_{ii} = \begin{cases} 1 & \text{if } i \text{ is odd} \\ 0 & \text{if } i \text{ is even} \end{cases}$$

The Roughness Penalty Matrix R is a sparse matrix assembled from local stiffness matrices.

Let the effective interval between point i and $i+1$ be:

$$\tau_i = (x_{i+1} - x_i) + \gamma_i.$$

For each interval $i = 1 \ldots n-1$, compute the local 4×4 matrix $K^{(i)}$:

$$K^{(i)} = \frac{1}{\tau_i^3} \begin{bmatrix} 12 & 6\tau_i & -12 & 6\tau_i \\ 6\tau_i & 4\tau_i^2 & -6\tau_i & 2\tau_i^2 \\ -12 & -6\tau_i & 12 & -6\tau_i \\ 6\tau_i & 2\tau_i^2 & -6\tau_i & 4\tau_i^2 \end{bmatrix}$$

Add $K^{(i)}$ into the global matrix R (size $2n \times 2n$) at the indices corresponding to points i and $i+1$.

- Rows/Cols of $K^{(i)}$ $(1, 2, 3, 4)$ map to Global Rows/Cols $(2i-1, 2i, 2i+1, 2i+2)$.
- Overlapping indices (e.g., node $i+1$) must be summed.

Solve the linear equation for z:

$$(S + \lambda R)z = y^*.$$

The output of this step is the updated f (odd elements of z) and f' (even elements of z).

Step 2: The Warping Update (γ-step) In this step, f and f' are fixed. We update γ to reduce the roughness energy.

The roughness energy on interval i is $E_i = \int (f'')^2 dt$. We need $\frac{\partial E_i}{\partial \gamma_i}$. Let:

- $\Delta f = f_{i+1} - f_i$
- Sum of slopes $S_{sl} = f'_i + f'_{i+1}$
- Product term $P_{sl} = (f'_i)^2 + f'_i f'_{i+1} + (f'_{i+1})^2$

The analytic gradient for interval i is:

$$\nabla_i = -36\frac{(\Delta f)^2}{\tau_i^4} + 24\frac{\Delta f \cdot S_{sl}}{\tau_i^3} - 4\frac{P_{sl}}{\tau_i^2}$$

Calculate the proposed update step:

$$\gamma_{temp} = \gamma_{current} - \eta \cdot \nabla$$

Projection onto the Simplex ($\mathcal{P}_M$). We must ensure $\gamma_{new} \geq 0$ and $\sum \gamma_{new} \leq M$. We apply the *Water-filling Algorithm*:

- Let $v = \gamma_{temp}$. Set any negative components to 0.
- If $\sum v \leq M$, return v.
- Sort v in descending order: $u_1 \geq u_2 \geq \dots$.
- Compute cumulative sum of u: $S_k = \sum_{j=1}^{k} u_j$.
- Find the index ρ (the number of active elements). It is the largest k such that:
$$u_k - \frac{S_k - M}{k} > 0$$
- Calculate the threshold θ:
$$\theta = \frac{S_\rho - M}{\rho}$$
- Final update: $\gamma_i = \max(v_i - \theta, 0)$.

Convergence and Post-Processing Repeat Steps 1 and 2 until the change in the variables falls below a tolerance level. To evaluate the model complexity (useful for GCV or AIC), calculate the trace of the smoothing matrix, which provides the effective degrees of freedom (EDF).

$$A = (S + \lambda R)$$

$$\text{EDF} = \text{Trace}(A^{-1}S)$$

Since S is diagonal with 1s and 0s, this is equivalent to summing the odd-indexed diagonal elements of A^{-1}.

References

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